AI-DRIVEN CLIMATE-AWARE DECISION SUPPORT AND RISK FORECASTING SYSTEMS FOR SRI LANKA

PROJECT ID : R26-IT-046



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AI-Driven Climate Risk & Decision Support for Sri Lanka

Problem

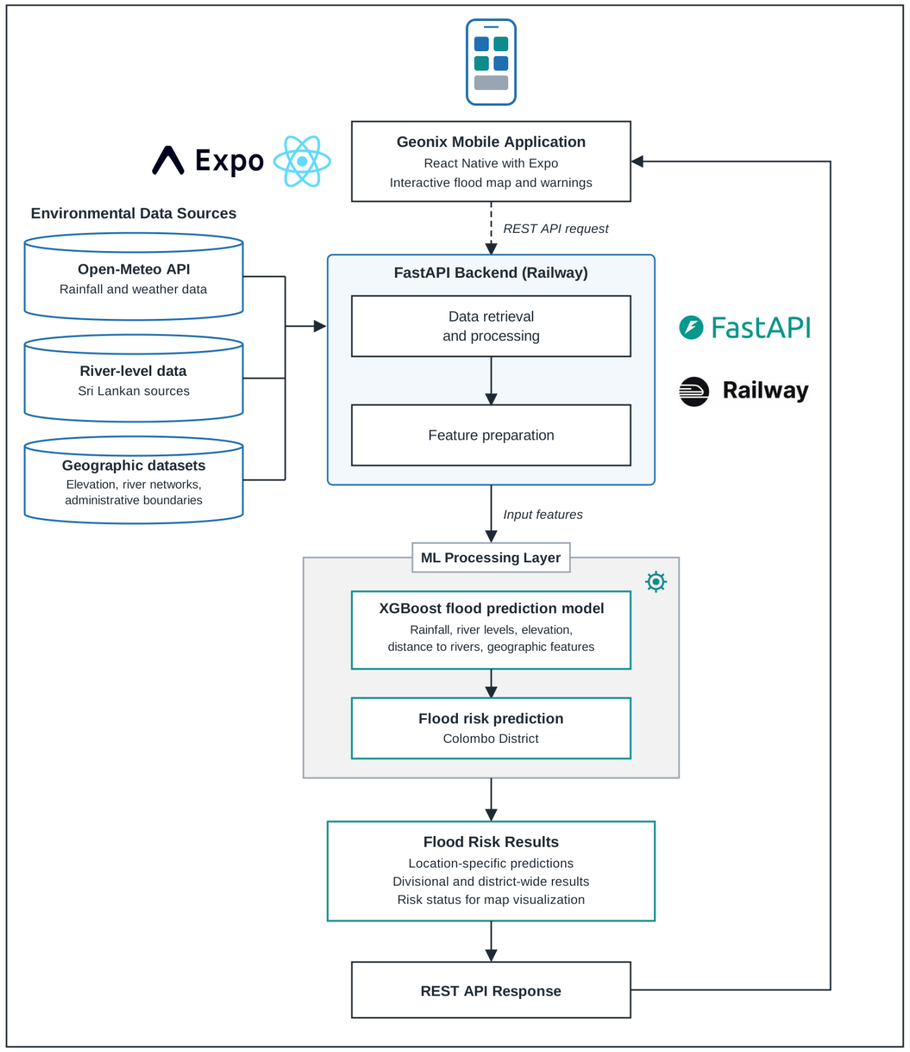
- People **do not receive** early flood warnings.
- People are **not aware** of dengue risk levels in their surrounding areas.
- Farmers do **not receive** specific advice related to weather and flood conditions.
- People find it difficult to identify **safe travel routes** during floods.

Design Excellence

- Easy-to-use interface for flood information access
- Clear interactive flood risk map
- Accurate predictions using trained ML models
- Real-time alerts with updated data

AI BASED FLOOD PREDICTION SYSTEM FOR EARLY DISASTER WARNING





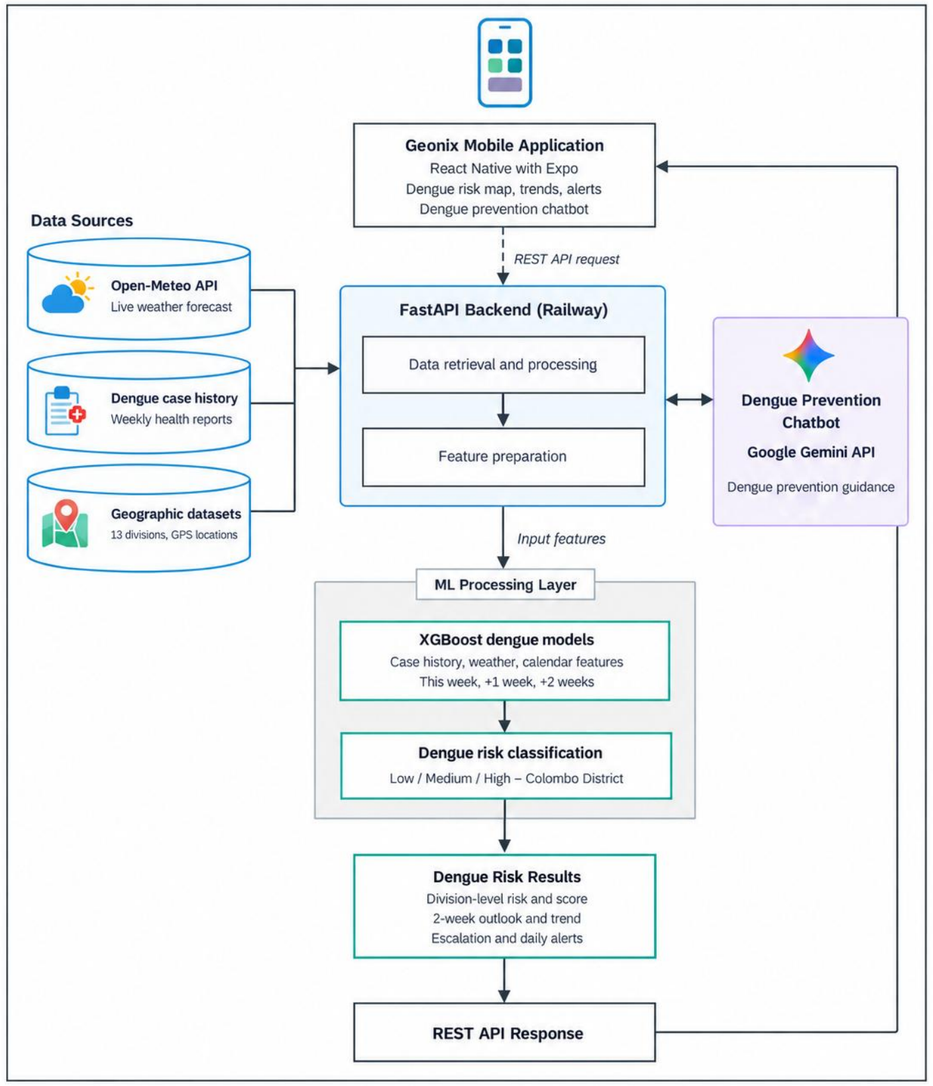
- **Flood Prediction Model:** Developed an XGBoost model to predict flood risk in Colombo District.
- **Real-Time Data Integration:** Integrated rainfall and river-level data for flood prediction.
- **Backend Development:** Built a FastAPI backend to process data and deliver predictions through REST APIs.
- **Flood Map Visualization:** Displayed location-based flood risks and warnings in the Geonix mobile app.

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AI-DRIVEN CLIMATE-BASED DENGUE RISK EARLY WARNING SYSTEM





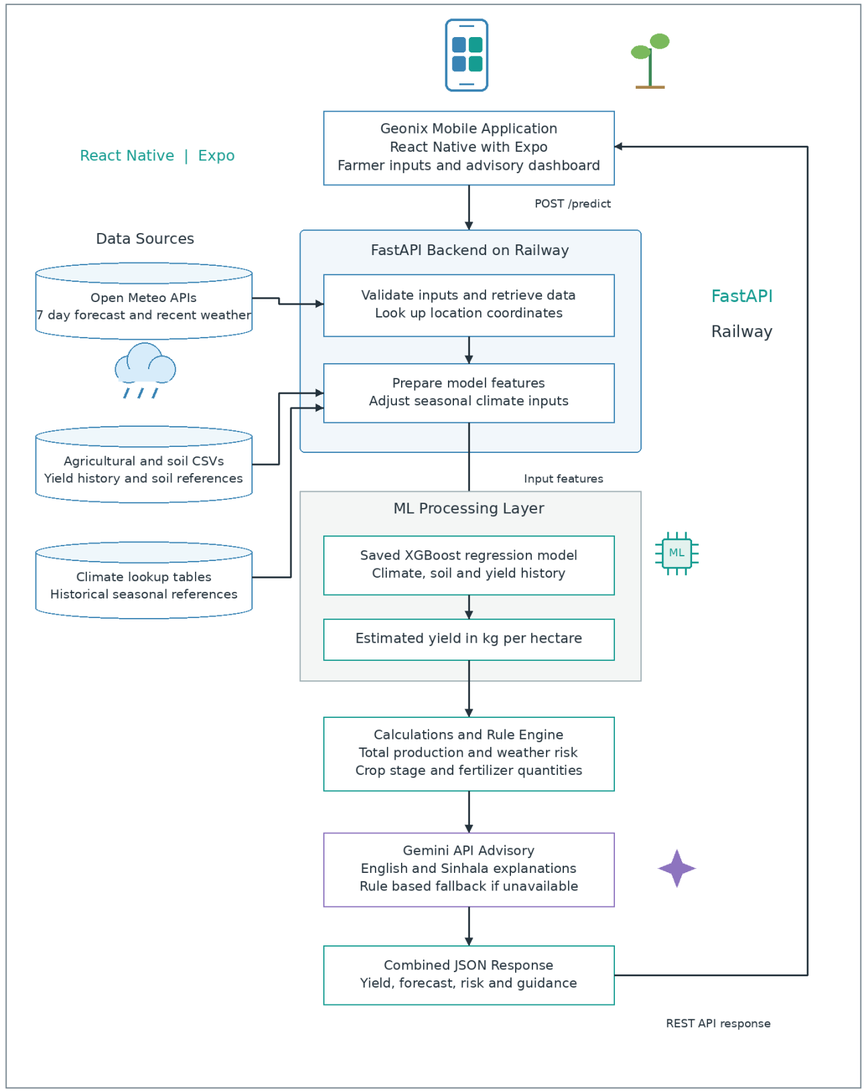
- **Dengue Risk Prediction:** Developed XGBoost models to predict dengue risk across 13 Colombo divisions.
- **Climate Data Integration:** Integrated historical dengue cases and weather forecasts.
- **Backend Development:** Built FastAPI REST APIs for dengue predictions.
- **Risk Visualization:** Displayed dengue risk maps, forecasts, trends, and alerts in Geonix mobile app.
- **AI Chatbot Integration:** Integrated Google Gemini API for dengue prevention guidance.

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CLIMATE-INFORMED PADDY ADVISORY SYSTEM





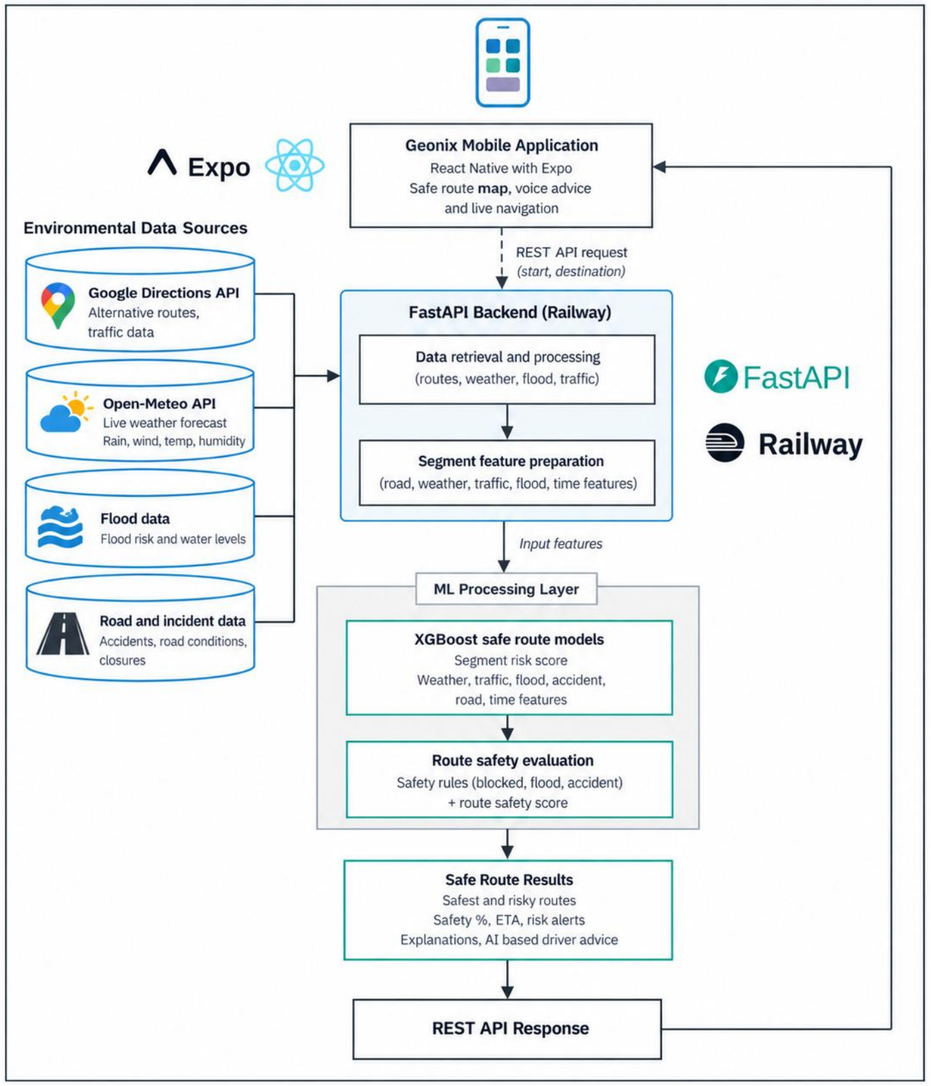
- **Yield Prediction:** Developed an XGBoost model to estimate paddy yield.
- **Weather Integration:** Integrated recent weather and forecasts using Open-Meteo APIs.
- **Backend Development:** Built a FastAPI backend for predictions and advisory processing.
- **Bilingual Dashboard:** Displayed yield, weather risks, and guidance in English and Sinhala.

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AI-DRIVEN CLIMATE-AWARE SAFE ROUTING SYSTEM





- **Safe Route Prediction:** Developed XGBoost models to evaluate road-segment safety risks.
- **Real-Time Data Integration:** Integrated traffic, weather, flood, and road incident data.
- **Backend Development:** Built FastAPI REST APIs to process data and generate safe routes.
- **Safe Route Visualization:** Displayed safest routes, ETAs, risk alerts, and AI-based driver advice in Geonix.

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Results

Name of the Models	Accuracy	F1 Score
Flood Prediction (XGBoost Classification)	0.89	0.60
Dengue Risk Early Warning (XGBoost Classification)	0.84	0.83
Paddy Advisory System (XGBoost Regression)	0.87	N/A (Regression Model)
Safe Routing System (XGBoost Classification)	0.81	0.83

LIMITATIONS

- The app is limited to the Colombo District only.
- The paddy model should provide recommendations based on different paddy varieties.
- The dengue model only predicts dengue risk, not other mosquito-borne diseases.

Future Improvements

- Expand the app to cover other districts.
- Add support for multiple paddy varieties.
- Extend dengue predictions to other mosquito-borne diseases.

THANK YOU .....

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